Summary and Recommendation

📌 Project Overview

This project analyzes the **Pizza Hut Sales dataset** to understand customer ordering patterns, pizza sales, revenue, product categories, pizza sizes, and order timing. The objective of the analysis is to identify important business patterns and gain insights into the company's sales performance and customer preferences.

The analysis was performed using **Python, Pandas, NumPy, Matplotlib, Seaborn, MySQL, and MySQL Connector**.

🎯 Project Objectives

The main objectives of this project were to:

- Understand the Pizza Hut dataset and its database structure.
- Import and manage the dataset using **MySQL**.
- Analyze the total number of customer orders.
- Calculate the total revenue generated.
- Identify the highest-priced pizza.
- Analyze pizza size distribution.
- Identify the most ordered pizzas.
- Analyze sales across different pizza categories.
- Understand customer ordering patterns throughout the day.
- Calculate the average number of pizzas ordered per day.

- Identify the top revenue-generating pizzas.
- Analyze revenue contribution by pizza category.
- Track cumulative revenue over time.
- Identify the top-performing pizzas within each category.

🧹 Data Preparation and Database Setup

The dataset consisted of four main tables:

- **Pizzas**
- **Orders**
- **Pizza Types**
- **Order Details**

The data was imported from CSV files into a **MySQL database**.

Dataset Records

The project successfully imported:

- **96 rows** into the `pizzas` table.
- **21,350 rows** into the `orders` table.
- **32 rows** into the `pizza_types` table.
- **48,620 rows** into the `order_details` table.

Python was then connected to the MySQL database using **MySQL Connector** to perform SQL queries and analyze the results.

📊 Exploratory Data Analysis

1. Total Number of Orders

The total number of orders was calculated from the orders table.

Analysis:

The dataset contains:

21,350 total orders.

Business Insight:

This provides an overview of the overall business activity and helps measure the volume of customer transactions.

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2. Total Revenue Analysis

Total revenue was calculated by multiplying the quantity of pizzas ordered by their respective prices.

Analysis:

The Pizza Hut dataset generated total revenue of approximately:

$817,860.05

Business Insight:

Revenue is one of the most important business metrics. It helps the company understand its overall sales performance.

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3. Highest-Priced Pizza

The pizzas were analyzed based on their prices to identify the most expensive pizza.

Analysis:

The Greek Pizza was identified as the highest-priced pizza.

Business Insight:

Premium-priced products can contribute significantly to revenue. The company can analyze whether high-priced pizzas generate sufficient demand and revenue.

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4. Pizza Size Distribution

The number of pizzas was analyzed according to their sizes.

Analysis:

The **Small (S)** size had the highest number of pizza varieties in the dataset.

Business Insight:

Analyzing pizza sizes helps understand the company's product offerings and can support inventory and menu planning.

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5. Most Ordered Pizzas

The pizzas were ranked based on the total quantity ordered.

Analysis:

Some of the most frequently ordered pizzas include:

- **The Classic Deluxe Pizza**
- **The Barbecue Chicken Pizza**
- **The Hawaiian Pizza**
- **The Pepperoni Pizza**
- **The Thai Chicken Pizza**

Among these, **The Classic Deluxe Pizza** had the highest quantity ordered.

Business Insight:

Popular pizzas should receive greater attention in:

- Inventory management
- Ingredient availability
- Menu promotions
- Customer recommendations

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6. Pizza Category Analysis

The number of pizzas ordered was analyzed across different categories.

The main categories were:

- **Classic**
- **Supreme**
- **Veggie**
- **Chicken**

Analysis:

The **Classic category** had the highest number of pizzas ordered.

The order quantities were:

- **Classic:** 14,888
- **Supreme:** 11,987
- **Veggie:** 11,649
- **Chicken:** 11,050

Business Insight:

The Classic category appears to be the most popular among customers. Pizza Hut can continue focusing on popular Classic pizzas while developing strategies to increase demand for other categories.

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7. Customer Ordering Pattern by Hour

The number of orders was analyzed according to the hour of the day.

Analysis:

Customer orders increased significantly during certain hours of the day.

The busiest periods included:

- **12 PM**
- **1 PM**
- **5 PM**

- **6 PM**

The highest number of orders occurred around **12 PM**.

Business Insight:

Understanding peak ordering hours can help Pizza Hut optimize:

- Staff scheduling
- Kitchen operations
- Delivery resources
- Inventory preparation

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8. Number of Pizza Types by Category

The number of pizza varieties available in each category was analyzed.

Analysis:

The dataset contains:

- **Chicken:** 6 pizza types
- **Classic:** 8 pizza types
- **Supreme:** 9 pizza types
- **Veggie:** 9 pizza types

Business Insight:

Supreme and Veggie categories have the highest number of pizza varieties. However, having more varieties does not necessarily guarantee higher sales.

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9. Average Pizzas Ordered Per Day

The average number of pizzas ordered per day was calculated.

Analysis:

On average, approximately:

138 pizzas were ordered per day.

Business Insight:

Daily demand analysis can help Pizza Hut estimate ingredient requirements and improve inventory planning.

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10. Top Revenue-Generating Pizzas

The pizzas were analyzed based on the total revenue generated.

Analysis:

The top three revenue-generating pizzas were:

1. **The Thai Chicken Pizza — $43,434.25**
2. **The Barbecue Chicken Pizza — $42,768.00**
3. **The California Chicken Pizza — $41,409.50**

Business Insight:

Interestingly, the pizzas generating the highest revenue are from the **Chicken category**. These products should be considered important revenue drivers.

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11. Revenue Contribution by Category

The percentage contribution of each category to total revenue was calculated.

Analysis:

Revenue contribution was approximately:

- **Classic:** 26.91%
- **Supreme:** 25.46%
- **Chicken:** 23.96%
- **Veggie:** 23.68%

Business Insight:

The Classic category contributed the highest percentage of total revenue. However, all four categories contributed relatively significant amounts to overall revenue.

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12. Cumulative Revenue Analysis

Cumulative revenue was calculated over time using SQL window functions.

Analysis:

The cumulative revenue increased throughout the year and reached approximately:

$817,860.05

by the end of the available data.

Business Insight:

Cumulative revenue helps track the company's financial growth over time and provides a clear picture of overall sales performance.

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13. Top Revenue-Generating Pizzas Within Each Category

The project used SQL ranking functions to identify the top three pizzas based on revenue within each category.

Analysis:

The analysis identified the strongest revenue-generating products in:

- Chicken
- Classic
- Supreme
- Veggie

For example:

Chicken Category:

- The Thai Chicken Pizza
- The Barbecue Chicken Pizza
- The California Chicken Pizza

Business Insight:

Analyzing top products within each category helps the company understand which specific products are driving category performance.

🔑 Key Insights

Based on the analysis, the following important insights were identified:

✅ The dataset contains **21,350 total orders**.

✅ Total revenue generated was approximately **$817,860.05**.

✅ **The Greek Pizza** was the highest-priced pizza.

✅ **The Classic Deluxe Pizza** was the most frequently ordered pizza.

✅ The **Classic category** had the highest number of pizzas ordered.

✅ The highest customer ordering activity occurred around **12 PM**, with another busy period during the evening.

✅ Approximately **138 pizzas were ordered per day on average**.

✅ **The Thai Chicken Pizza** generated the highest revenue.

✅ The **Classic category contributed the highest percentage of total revenue**.

✅ SQL window functions were used to calculate **cumulative revenue** and rank top pizzas within each category.

💡 Business Recommendations

1. Maintain Inventory for Popular Pizzas

The company should ensure sufficient availability of highly ordered pizzas such as:

- The Classic Deluxe Pizza
- The Barbecue Chicken Pizza
- The Hawaiian Pizza
- The Pepperoni Pizza

This can help prevent stock shortages during high-demand periods.

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2. Focus on High-Revenue Pizzas

The **Thai Chicken Pizza**, **Barbecue Chicken Pizza**, and **California Chicken Pizza** generate significant revenue.

Pizza Hut should consider:

- Promoting these pizzas
- Maintaining ingredient availability
- Including them in targeted marketing campaigns

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3. Optimize Operations During Peak Hours

Since customer orders are highest around lunchtime and evening hours, Pizza Hut should increase operational capacity during these periods.

This may include:

- Better staff scheduling
- Faster food preparation
- Improved delivery management

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4. Strengthen the Classic Category

The Classic category has both the highest order quantity and the largest revenue contribution.

The company should continue investing in this category through:

- Promotions
- Menu visibility
- Product recommendations

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5. Improve Lower-Performing Categories

Categories such as Veggie and Chicken have lower total order quantities than Classic.

The company can investigate whether promotions, pricing strategies, or additional marketing could increase demand.

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6. Use Product-Level Performance for Marketing

Pizza Hut should not only analyze broad categories. Individual pizza performance should also be considered.

High-performing pizzas within each category can be promoted, while lower-performing products can be reviewed.

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7. Use Daily Demand for Inventory Planning

With an average of approximately **138 pizzas ordered per day**, demand patterns can help improve:

- Ingredient purchasing
- Inventory management
- Kitchen preparation

This can reduce waste and improve operational efficiency.

📌 Conclusion

This **Pizza Hut Sales Analysis project** explored important business metrics including **total orders, total revenue, pizza prices, pizza sizes, product categories, customer ordering patterns, daily demand, revenue contribution, and top-performing pizzas**.

The project demonstrates the integration of **MySQL and Python** for data analysis. SQL was used to extract and analyze business data, while **Pandas, Matplotlib, Seaborn, and NumPy** were used for data handling and visualization.

Overall, the analysis provides useful insights into **customer preferences, popular products, revenue drivers, category performance, and peak ordering times**. These insights can help Pizza Hut make better decisions related to **inventory management, staffing, marketing strategies, menu planning, and sales optimization**.

This is a strong SQL + Python Data Analytics project, especially because it demonstrates database integration, joins, aggregation, grouping, and SQL window functions.